

Essential Summary

Quantitative Analysis of Dynamical Bifurcations in a Coupled Smooth and Discontinuous Oscillator with High-order Nonlinear Damping

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1 Background

SD L-P GHFP MGHFP
SD

The coupled smooth and discontinuous (SD) oscillator involves two irrational nonlinear terms in its restoring force, which severely hinders quantitative analysis. Classical perturbation methods cannot handle this system directly. This work extends the MGHFP method to a coupled SD oscillator with high-order nonlinear damping.

2 Core Theory

2.1 System Model

SD

$$\ddot{x} + (x + \beta) \frac{1}{\sqrt{(x + \beta)^2 + \alpha^2}} + (x - \beta) \frac{1}{\sqrt{(x - \beta)^2 + \alpha^2}} = (\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4) \dot{x} \quad (1)$$

$\alpha \quad \alpha = 0 \quad \beta \quad \mu_c \quad \mu_1 - \mu_4$

2.2 MGHFP Key Steps of MGHFP

1. $\frac{d\varphi}{dt} = \Phi(\varphi)$ φ
2. **Fourier** $x(t) = a \cos^2 \Phi(t) + b$ ε
 $a = a_0 + \varepsilon a_1 + \varepsilon^2 a_2 + \dots, \quad \Phi = \Phi_0 + \varepsilon \Phi_1 + \varepsilon^2 \Phi_2 + \dots \quad (2)$
3. **Simpson** — Simpson Fourier p_{2i}

3 Key Equations

3.1 Amplitude-Parameter Relation

Fourier $m = 4$ A μ_c

$$\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)}(E_1 \mu_1 + E_2 \mu_2 + E_3 \mu_3 + E_4 \mu_4) \quad (3)$$

p_{2i} Fourier Simpson E_i a_0 b p_{2i} —

3.2 Stability Criterion

h_0

$$h_0 = \frac{1}{\pi} \int_0^\pi \left[\mu_c + \sum_{i=1}^4 \mu_i (a_0 \cos^2 \varphi + b)^i \right] d\varphi \quad (4)$$

- $h_0 > 0$ (unstable)
- $h_0 = 0$ (semi-stable)
- $h_0 < 0$ (stable)

3.3 Homo-heteroclinic Bifurcation

$$(3) \quad b = h \quad \mu_c^{\text{hom}} \quad b = h_2 \quad a_0 + b = h_1 \quad \mu_c^{\text{het}}$$

3.4 Approximate Solution

$$x = 2A \cos^2 \Phi - A + \mathcal{O}(\varepsilon^2), \quad \frac{d\varphi}{dt} = \Phi_0(\varphi) + \varepsilon \Phi_1(\varphi) + \mathcal{O}(\varepsilon^2) \quad (5)$$

4 Key Findings

1. MGHP SD —
2. h_0
3. SD MGHP Runge–Kutta
4. $\rightarrow + \rightarrow$
5. $|\varepsilon \mu_i| \leq 0.5$ Fourier

5 Limitations

- $\varepsilon > 0.5$ 5%
- Fourier
- $\alpha = 0$ ε

6 Applications

- (1) QZS (2) MEMS /